

1 Summary

Model (Run-name): model1 (test)

Iteration: 14

TAC: 154693.69914866655

Took: 1.5339586734771729s

Total Time: 18.558789253234863s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

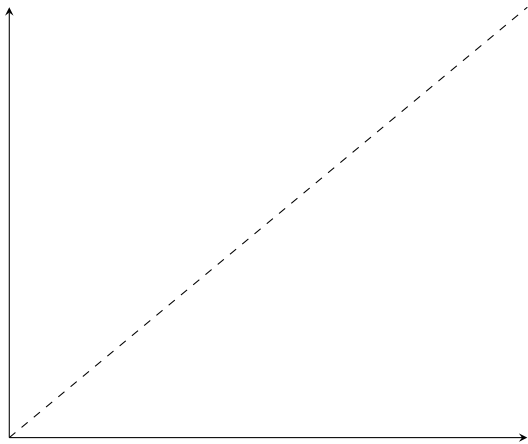
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

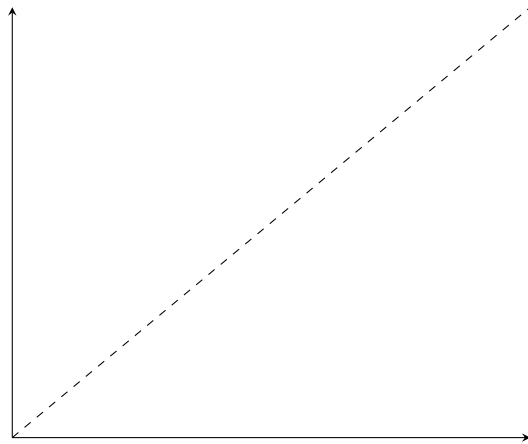
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05145929433939766	$b_{1,1,1}^{H,out}$	930.9532115792922
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.018526534234860233	$b_{1,2,2}^{H,out}$	3845.494866912968
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02710276238507339	$b_{2,1,2}^{H,out}$	9406.757699630549
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014871535343152149	$b_{1,1,1}^{C,in}$	8543.242300369451
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.00951467090778529	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022385804428626338	$b_{2,1,2}^{C,in}$	4812.422226926419
$A_{1,1,1}^\beta$	71.88123217857378	$q_{1,1,1}$	704.5051330870319	$b_{1,1,1}^{C,out}$	9247.747433456483
$A_{1,2,2}^\beta$	72.25348351595485	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	128.52021081477858	$q_{2,1,2}$	2393.2423003694516	$b_{2,1,2}^{C,out}$	7205.664527295869
$A_{cu,1}^\beta$	3.9948823402911255	$q_{cu,1}$	145.49486691296784		
$A_{cu,2}^\beta$	38.10872593446926	$q_{cu,2}$	2006.7576996305484		
$A_{hu,1}^\beta$	13.439500729482463	$q_{hu,1}$	502.2525665435164		
$A_{1,1,1}$	71.88123217857378	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	72.25348351595485	$t_{1,2}^{(H)}$	579.5494866912968		
$A_{2,1,2}$	128.52021081477858	$t_{1,3}^{(H)}$	384.5494866912968		
$A_{cu,1}$	3.9948823402911255	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	38.10872593446926	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.439500729482463	$t_{2,3}^{(H)}$	470.3378849815274		
$dt_{1,1,1}$	33.48350443623441	$t_{1,1}^{(C)}$	616.5164955637656		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	569.5494866912967		
$dt_{1,2,2}$	79.5494866912968	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	34.54948669129678	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	20.450513308703307	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	60.337884981527424	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	64.54948669129678	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	150.33788498152742	$t_{1,2,2}^H$	384.5494866912968		
$dt_{hu,1}$	63.483504436234426	$t_{2,1,2}^H$	470.33788498152745		
$f_{1,1,1}^H$	2.516089761025114	$t_{1,1,1}^C$	616.5164955637655		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	606.523181231542		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	1635.4583446663241		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5795.494866912968		
$f_{2,1,2}^C$	11.737615187625412	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

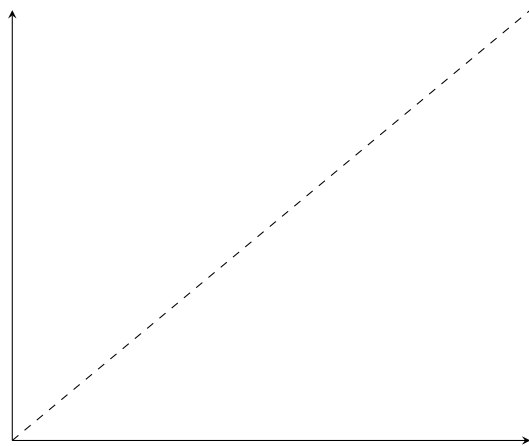
- $(1, 1, 1)$
 - Value: 0.051459
 - Error: 0.000000990174728
 - Relative error: 0.000019241532324
- $(1, 2, 2)$
 - Value: 0.018527
 - Error: 0.000006501520384
 - Relative error: 0.000350807092233
- $(2, 1, 2)$
 - Value: 0.027103
 - Error: 0.000022419175289
 - Relative error: 0.000826507842521
- $cu, 1$
 - Value: 0.014872
 - Error: 0.000001022947629
 - Relative error: 0.000068780878793
- $cu, 2$
 - Value: 0.009515
 - Error: 0.000000019272839
 - Relative error: 0.000002025587647
- $hu, 1$
 - Value: 0.022386
 - Error: 0.000000822373222
 - Relative error: 0.000036735021716



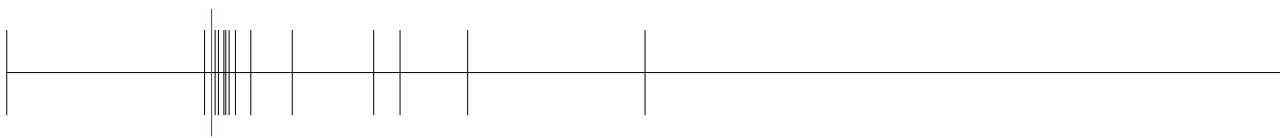
(a) *RecLMTD* - (1, 1, 1)



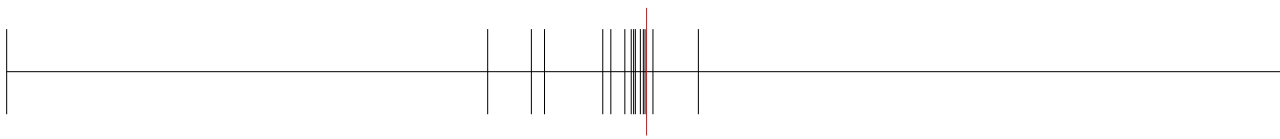
(b) *RecLMTD* - (1, 2, 2)



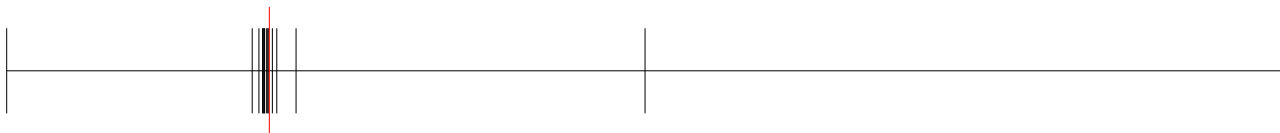
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1635.4583446663241
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5795.494866912968
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 930.9532115792922
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3845.494866912968
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9406.757699630549
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8543.242300369451
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4812.422226926419
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9247.747433456483
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 7205.664527295869
 - Absolute error: 86.52882362564196
 - Relative error: 0.012154400088346756



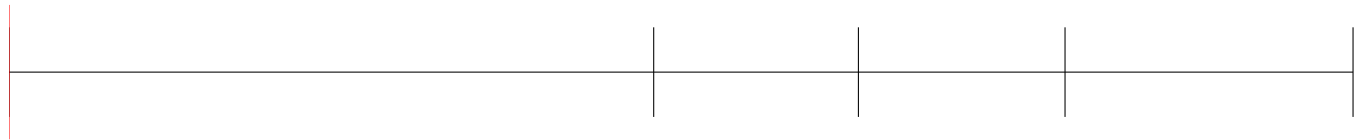
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



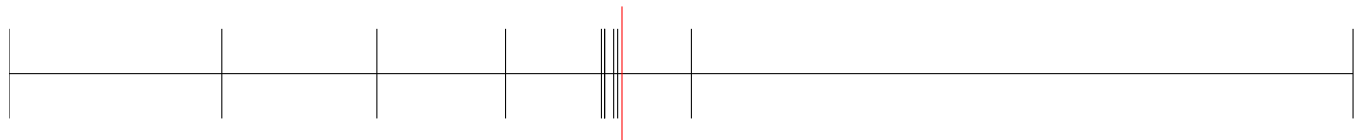
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



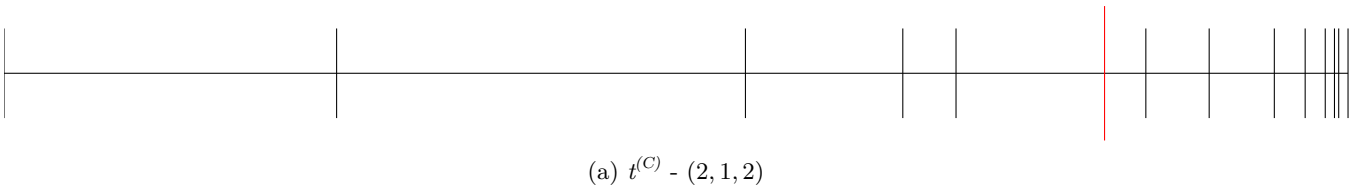
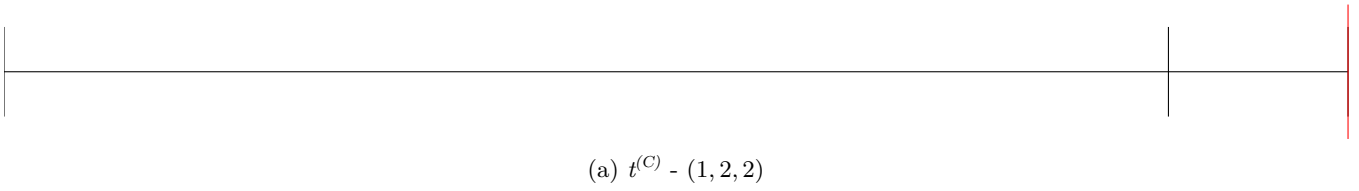
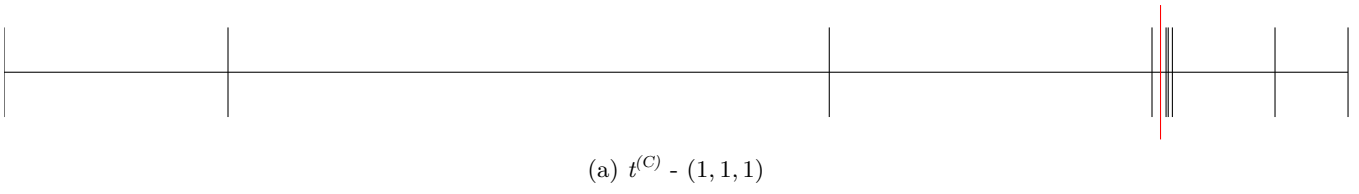
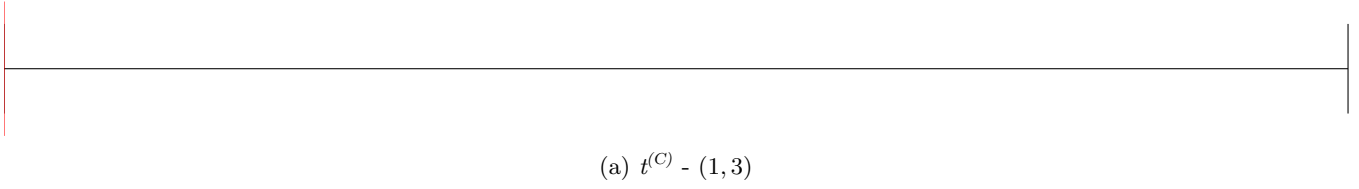
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

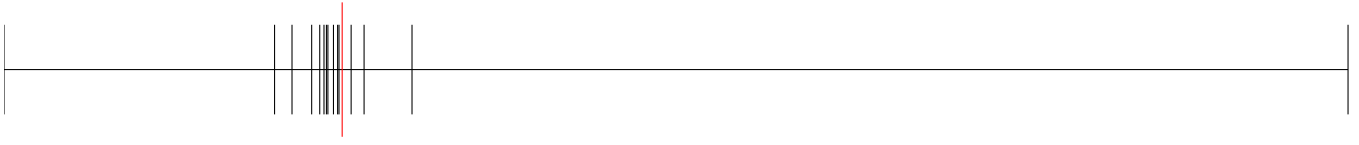
- $(1, 1, 1)$
 - Value: 71.881232
 - Error: 0.625441835710404
 - Relative error: 0.008625989872149
- $(1, 2, 2)$
 - Value: 72.253484
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 128.520211
 - Error: 1.206743978860800
 - Relative error: 0.009302183811995

5.2 A_{CU}

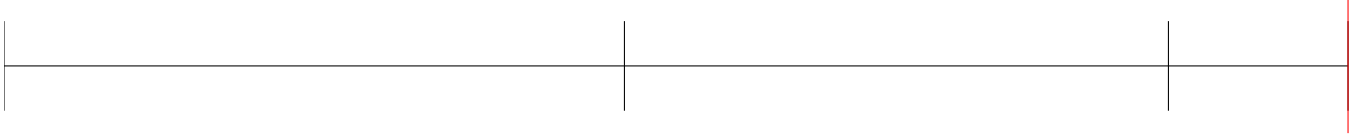
- $cu, 1$
 - Value: 3.994882
 - Error: 0.332581770795713
 - Relative error: 0.076853732869476
- $cu, 2$
 - Value: 38.108726
 - Error: 0.078552272828560
 - Relative error: 0.002057027274951

5.3 A_{HU}

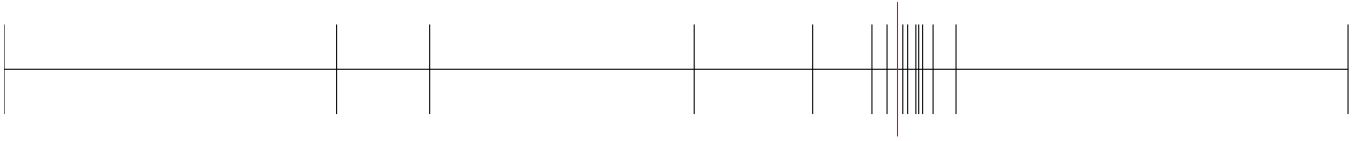
- $hu, 1$
 - Value: 13.439501
 - Error: 0.052492544620090
 - Relative error: 0.003890644143801



(a) $A^\beta - (1, 1, 1)$



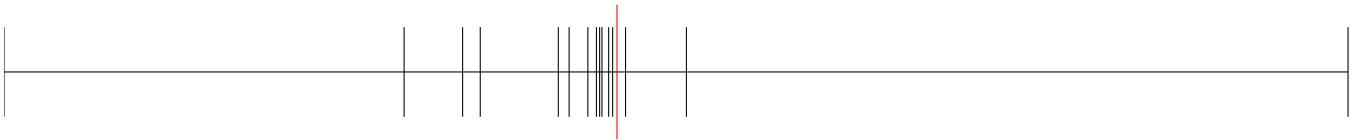
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

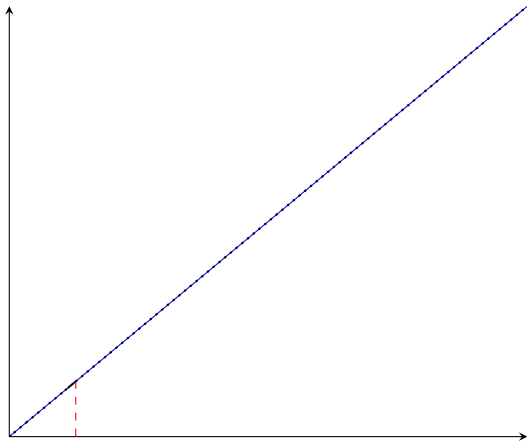


(a) $A^\beta - hu, 1$

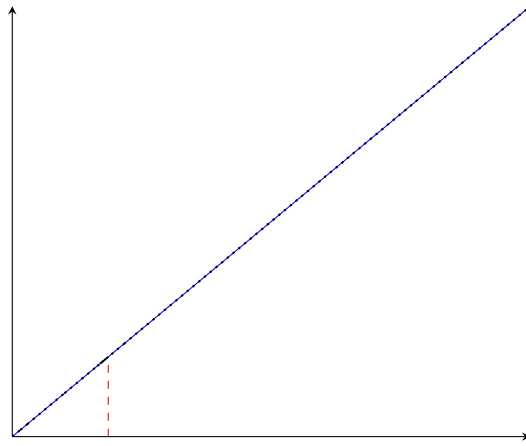
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.881232
 - $\hat{q}^\beta$: 71.881232
 - q^β : 71.881232
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 71.22933662920877, **71.6637831223933**, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 72.253484
 - $\hat{q}^\beta$: 72.253484
 - q^β : 72.253484
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, 69.34502840170047, **70.94081287188642**, **74.6496383881616**, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 128.520211
 - $\hat{q}^\beta$: 128.520211
 - q^β : 128.520211
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, **118.88765652962957**, **131.69760457629607**, 132.12228195956155, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 3.994882
 - $\hat{q}^\beta$: 3.994882
 - q^β : 3.994882
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, **3.1809419553840232**, **4.269280092716329**, 4.432280503220532, 4.983938626310074, 4.993068227518925, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 38.108726

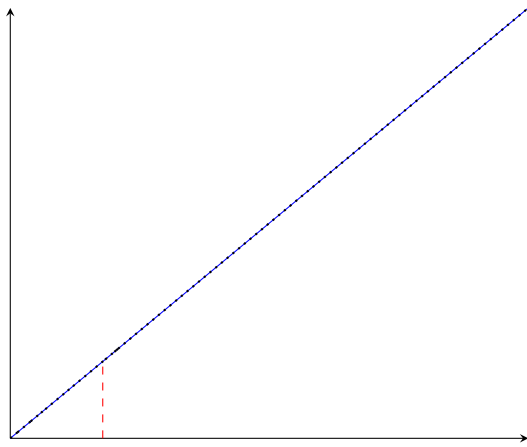
- $\hat{q}^\beta$: 38.108726
- q^β : 38.108726
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, **37.9454585315849**, **38.21652191748154**, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.439501
 - $\hat{q}^\beta$: 13.439501
 - q^β : 13.439501
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.192252946941991, 13.22717920580809, 13.244985069124171, 13.30871858348944, **13.346967780023082**, **13.468285806082731**, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



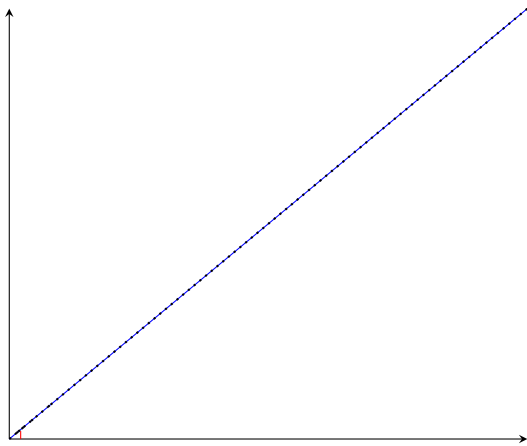
(a) $q^\beta - (1, 1, 1)$



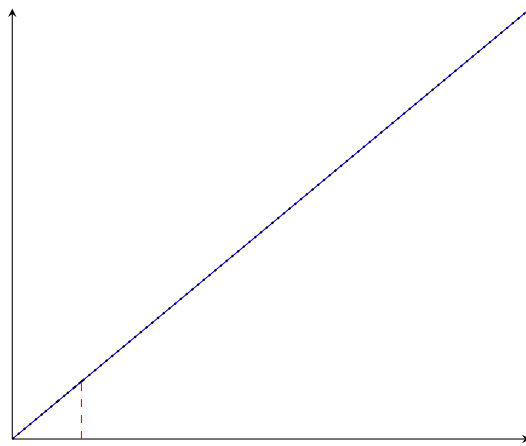
(b) $q^\beta - (1, 2, 2)$



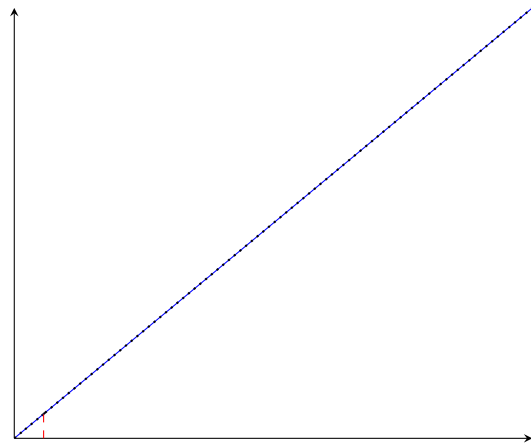
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1337.577773073581
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008125921192688932	$b_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	1337.5777730735817
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	418.8196134269517		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	370.0		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	575.1418898496516		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4864.541655333676		
$f_{1,2,1}^H$	7.483910238974886	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	3.262384812374588	$b_{1,2,1}^{H,out}$	4864.541655333676		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		